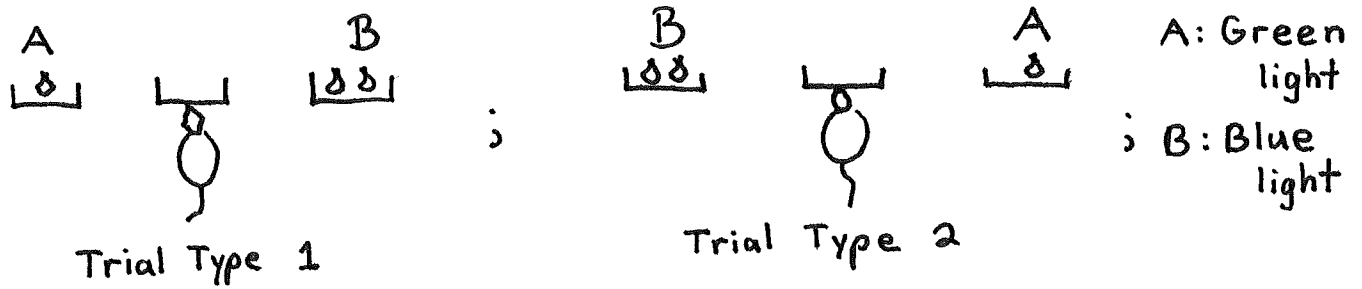


Learned Decisions:

Many real world decisions require learning.

Foraging: Am I more likely to find food I want at Trader Joe's or Jewel? Is a rat more likely to find food in a blue or green garbage bin?

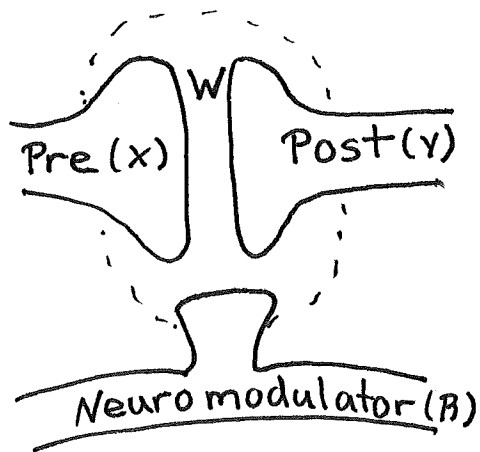


A or B?
What are statistics of this decision?

$P(R|A) = P_A, P(R|B) = P_B$
Experimentally Set, arbitrary

This behavior requires learned associations between stimulus (color of light) and reward (amount of water).

Three-factor plasticity



$$\Delta W = \Delta W(x, y, R)$$

R: Reward

Neuromodulator canonically associated with reward learning is Dopamine.

Other canonical learned decisions:

- Navigation: How do I find the water source?
- Economic valuation: How much will I pay for brand-name shampoo?

Covariance-based plasticity:

Dopamine speeds up plasticity, and the simplest models treat it as a gain parameter.

$$\Delta W = \underbrace{f(x, y)}_{\text{e.g. Hebbian}} \underbrace{R}_{\text{like a learning rate}}$$

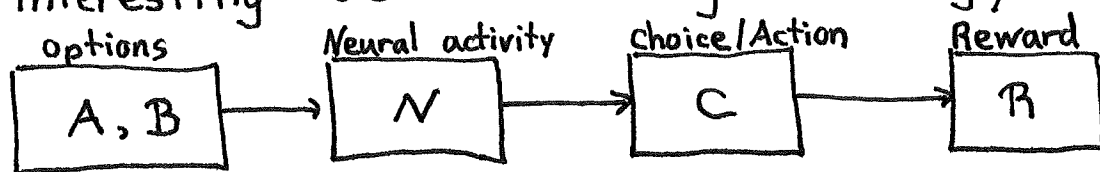
This can over-strengthen the synapse when reward keeps coming. Stop learning when reward is expected:

$$\Delta W = f(x, y)(R - \langle R \rangle)$$

Note that

$$\langle \Delta W \rangle = \langle f(x, y)(R - \langle R \rangle) \rangle = \text{Cov}(\underbrace{f(x, y)}_{\text{"N"}}, R)$$

so this is again a "covariance rule." Loewenstein & Seung showed that such plasticity rules lead to an interesting decision-making strategy.



$$\langle \Delta W \rangle = 0 \iff \text{Cov}(N, R) = 0 \iff \langle R \rangle_{R|C=A} = \langle R \rangle_{R|C=B}$$

The first statement follows directly from plasticity rule.

The second statement follows from

$$\begin{aligned} \text{Cov}(N, R) &= \langle (N - \langle N \rangle) R \rangle = \langle \langle (N - \langle N \rangle) R \rangle_{N, R|C} \rangle_C \\ &= P(C=A) \underbrace{\langle \delta N \rangle_{N|C=A} \langle R \rangle_{R|C=A}}_{\text{Reward depends on } N \text{ only through choice}} + P(C=B) \langle \delta N \rangle_{N|C=B} \langle R \rangle_{R|C=B} \end{aligned}$$

$$= P(C=A) \langle \delta N \rangle_{N|C=A} (\langle R \rangle_{R|C=A} - \langle R \rangle_{R|C=B})$$

The condition follows if we assume that the mean neural activity depends on choice (i.e. $\langle \delta N \rangle_{N|C=A} \neq 0$).

Matching behavior:

The condition that

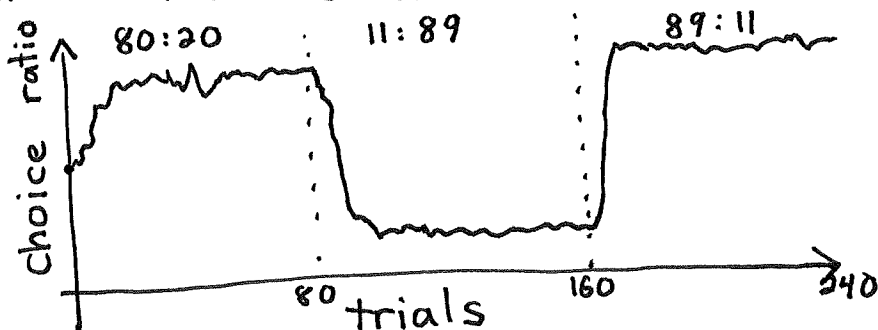
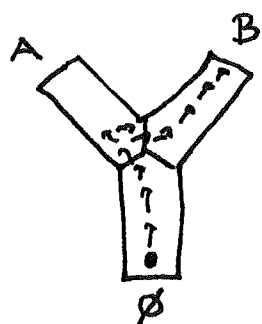
$$\langle R \rangle_{R|C=A} = \langle R \rangle_{R|C=B}$$

is known as "matching", because it implies that the animal matches its choice ratio to the reward ratio

$$\frac{R_A}{C_A} = \frac{R_B}{C_B} \Rightarrow \frac{C_A}{C_B} = \frac{R_A}{R_B}$$

$R_{A/B}$: Total rewards from options
 $C_{A/B}$: Total choices (by Herrnstein)

Matching was first described in pigeons and has since been found in humans, monkeys, mice, flies, and more. My collaborators and I discovered matching in flies and built a neural network model for it.



Computing choice and reward ratios over many blocks and flies revealed matching. Our neural network model was of the mushroom body.



Kenyon cells



Mushroom body output neurons



$$y = \sum_{i=1}^N W_i X_i$$

$$P(c = \text{"Approach"}) = \frac{e^{-\beta(y - \alpha)}}{1 + e^{-\beta(y - \alpha)}}$$

β : controls stochasticity of choice

α : biases choice

$$\Delta W_i = -\gamma X_i (R - \langle R \rangle)$$

Model needed $\langle R \rangle$ term to generate matching and fit sequential decisions of flies.

Predicting rewards:

Many models posit that animals predict rewards through a "value function". In the simplest case,

$$\underbrace{V}_{\text{Value}} = \sum_{i=1}^N \underbrace{W_i}_{\text{weighting coefficients}} \underbrace{X_i}_{\text{stimuli}}$$

The value is meant to predict rewards. Minimizing $(R-V)^2$ with gradient descent gives the Rescola-Wagner rule.

$$\Delta W_i = \eta X_i (R - V)$$

This rule predicts many results in classical conditioning. For example, Pavlovian associations occur when X predicts R through V . Blocking occurs because if S_1 predicts R , then there is no learning signal for the new associations of $S_1 + S_2 \rightarrow R$, so $S_2 \rightarrow \emptyset$.

Temporal difference (TD) learning is designed to predict rewards that require time for delivery. For instance, to navigate animals learn a sequence of actions that eventually lead to reward, and each decision is usually not accompanied by immediate reward. Want to learn an estimate of total future reward

$$\begin{aligned} R(t) &= R(t) + \gamma R(t+1) + \gamma^2 R(t+2) + \dots \\ &= R(t) + \underbrace{\gamma}_{\text{Temporal discounting}} R(t+1) \end{aligned}$$

An ideal value function would report

$$V(t) = \langle R(t) \rangle$$

$$R(t) = V(t) - \gamma V(t+1)$$

So the relevant error signal is

$$\delta(t) = R(t) - V(t) + \gamma V(t+1)$$

We again predict value through stimuli,

$$V(t) = \underbrace{\sum_{t'=0}^t \sum_{i=1}^N W_i(t') X_i(t-t')}_{\text{Discrete time convolution}}$$

and update weights as

$$\Delta W_i(t') = \gamma \delta(t) X_i(t-t')$$

Note that V depends on stimuli and state (e.g., how long a stimulus has been present). Denote this combination as Δ_t . It's often convenient to seek a value function that depends also on chosen actions.

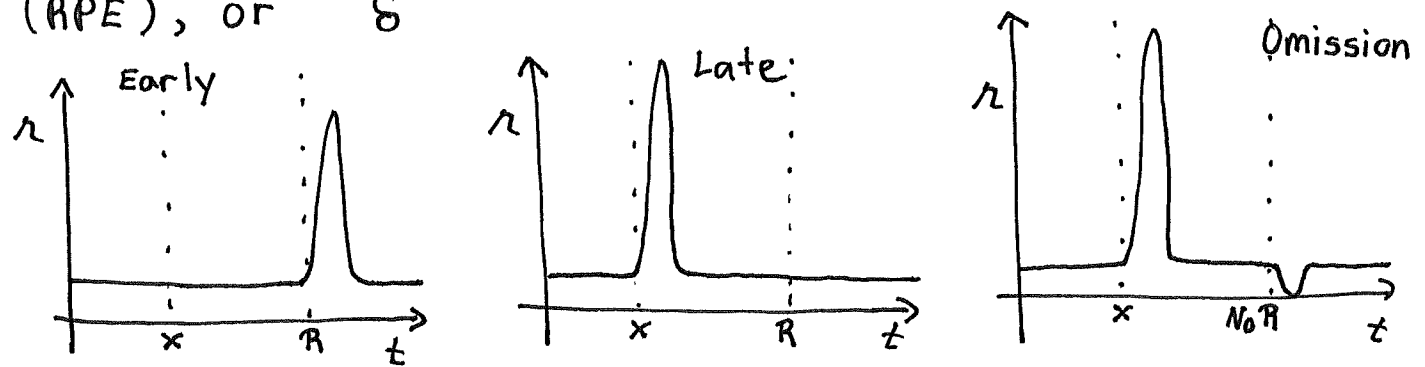
Q-learning: Seek to learn

$$Q(\underbrace{\Delta}_\text{state}, \underbrace{a}_\text{action}; t) = \langle R(t) \rangle_{R(t), R(t+1), \dots | \Delta_t = \Delta, a_t = a}$$

Q is called the state-action value function, or simply the "Q-value."

Neurobiology of RPE & Value:

Schultz, Dayan, and Montague famously argued that dopamine neuron activity resembles the reward prediction error (RPE), or δ



These dopamine neurons project to striatum. Subsequent work has found neural activity correlated to $V(t)$ and/or $Q(\Delta, a; t)$ in striatum. Such signals could support action selection. For instance

$$a_t = \arg \max_a Q(\Delta, a; t)$$